DATASHEET

# Ambient Light Sensor
# Surface - Mount
# ALS-PT19-315C/L177/TR8

## Features



- Light to Current, analog output
- Good output linearity across wide illumination range
- Guaranteed temperature performance, -40ºC to 85ºC
- Wide supply voltage range, 2.5V to 5.5V
- Size : 1.7mm(L)*0.8mm(W)*0.6mm(H)
- The product itself will remain within RoHS compliant version
- Compliance with EU REACH
- Compliance Halogen Free(Br < 900ppm, Cl < 900ppm, Br+Cl < 1500ppm)

## Description

The ALS-PT19-315C/L177/TR8 is a low cost ambient light sensor, consisting of phototransistor in miniature SMD. EVERLIGHT ambient light sensors are a good effective solution for power saving of display backlighting of mobile appliances, such as the mobile phones, NB and PDAs. Due to the high rejection ratio of infrared radiation, the spectral response of the ambient light sensor is close to that of human eyes.

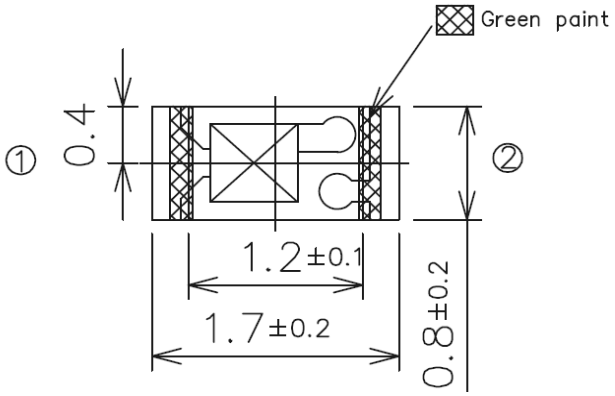
## Applications

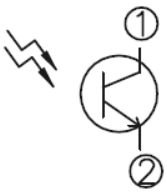
- Mobile devices – mobile phones, PDAs
- TVs, monitor
- Cameras
- Signage boards
- Street lighting
- CCD/CCTV
- Automatic residential and commercial management systems


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## Package Dimensions

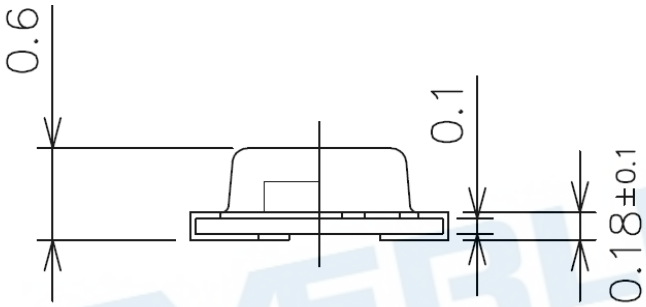
### Top View



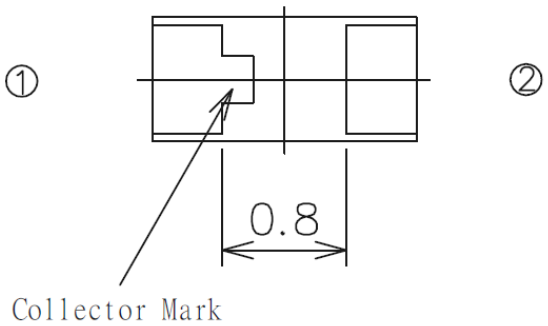




① Collector
② Emitter

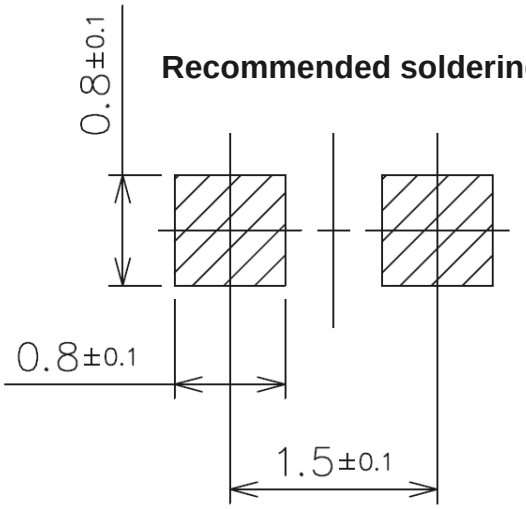
### Side View




### Bottom View




### Recommended soldering pad




**Unit: mm**
**Tolerances: ±0.1mm**


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## Absolute Maximum Ratings

| Parameter | Symbol | Rating | Unit |
|---|---|---|---|
| Supply Voltage | $V_{CC}$ | -0.5~6.0 | V |
| Operating Temperature Range | $T_{OPR}$ | -40 ~ +85 | °C |
| Storage Temperature Range | $T_{stg}$ | -40 ~ +100 | °C |
| Soldering Temperature [Note1] | $T_{sol}$ | 260 ± 10 | °C |

**Note1:** For detail reflow time and the recommended temperature profile, please refer to page 8.

## Recommended Operating Conditions

| Parameter | Symbol | Min. | Max. | Unit |
|---|---|---|---|---|
| Operating Temperature | $T_{OPR}$ | -40 | +85 | °C |
| Supply Voltage | $V_{CC}$ | 2.5 | 5.5 | V |

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## Electrical and Optical Characteristics ($T_a$=25℃ )

| Parameter | Symbol | MIN. | TYP. | MAX. | Unit | Test Condition |
|---|---|---|---|---|---|---|
| Dark Current | $I_{CEO}$ | --- | --- | 0.1 | uA | $V_{CE}$=10V, $E_v$= 0Lux |
| Collector-Emitter Saturation Voltage | $V_{CE(sat)}$ | --- | --- | 0.4 | V | $E_v$= 1000Lux |
| Light Current | $I_{PH1}$ | 5 | 15 | --- | uA | $V_{CE}$=5V, $E_v$= 100Lx [Note2] |
| | $I_{PH2}$ | 50 | 150 | --- | uA | $V_{CE}$=5V, $E_v$= 1000Lx [Note2] |
| | $I_{PH3}$ | 140 | 520 | --- | uA | $V_{CE}$=5V, $E_v$= 1000Lx [Note3] |
| Photocurrent Ratio | $I_{PH3}$ / $I_{PH2}$ | --- | 3.5 | --- | --- | $V_{CE}$=5V, $E_v$= 1000Lx |
| Saturation Output Voltage | $V_o$ | 4.5 | 4.6 | --- | V | $V_{CC}$=5V, $E_v$= 1000Lx[Note3] $R_L$=75K |
| Peak Sensitivity Wavelength | $\lambda_p$ | --- | 630 | --- | nm | --- |
| Sensitivity Wavelength Range | $\Lambda$ | 390 | --- | 700 | nm | --- |
| Rise time | $t_r$ | --- | 0.11 | --- | ms | $V_{CC}$ = 5 V, $R_L$ = 7.5KΩ |
| Fall time | $t_f$ | --- | 0.22 | --- | ms | |

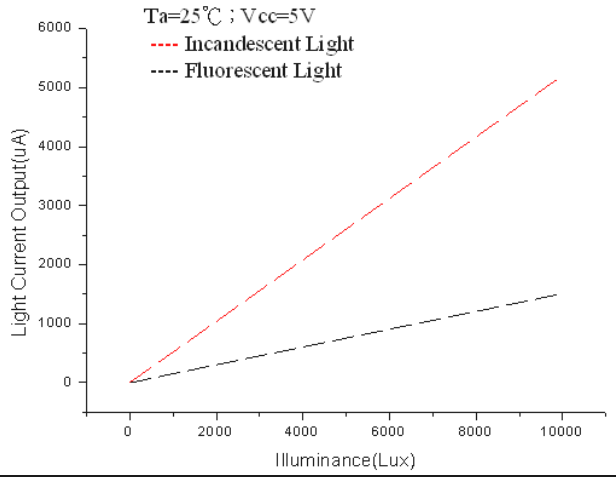
Note:

2. White Fluorescent light (Color Temperature = 6500K) is used as light source. However, White LED is substituted in mass production.
3. Illuminance by CIE standard illuminant-A / 2856K, incandescent lamp.

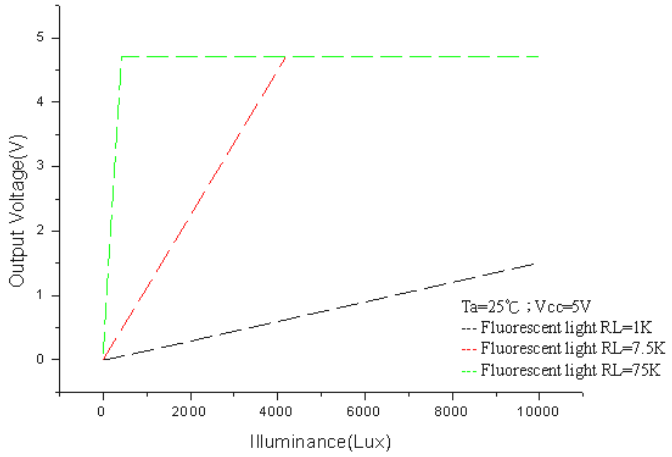

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## Typical Electrical and Optical Characteristics Curves

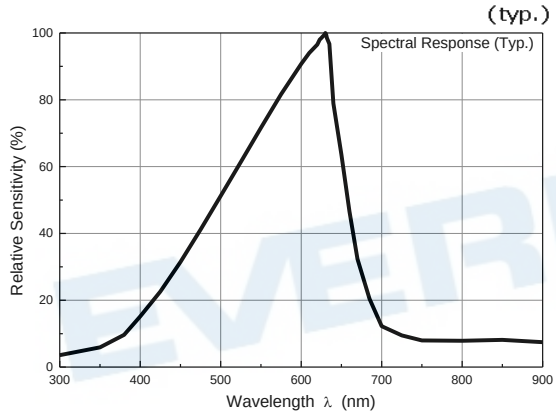
### Fig.1 Light Current vs. Illuminance



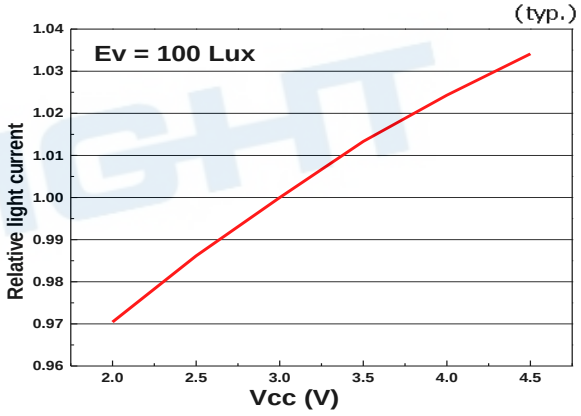

### Fig.2 Output Voltage vs. Illuminance



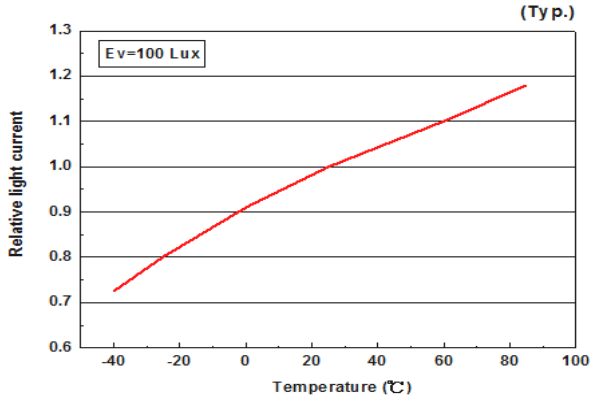

### Fig.3 Spectral Response



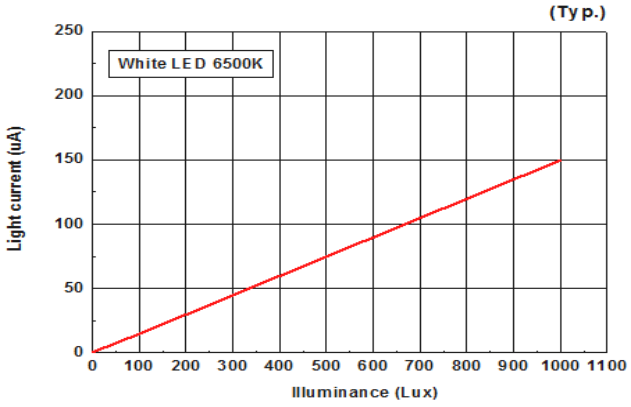

### Fig.4 Light current vs. Supply Voltage




### Fig.5 Light Current vs. Temperature



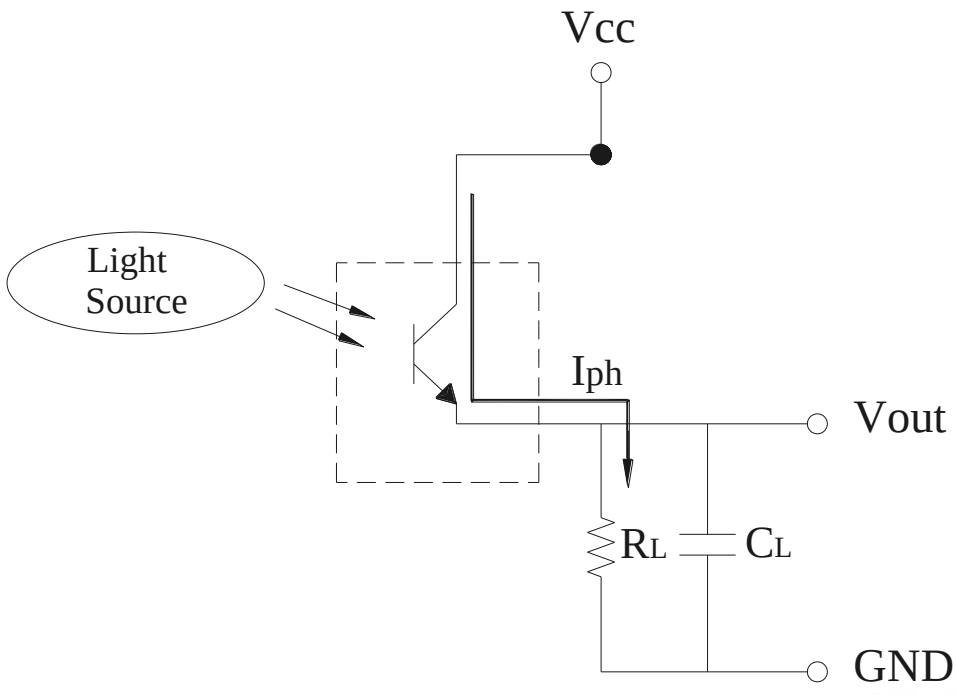

### Fig.6 Light Current vs. illuminance





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## Converting Photocurrent to Voltage




**Note:**

1. The output voltage ($V_{out}$) is the product of photocurrent ($I_{PH}$) and loading resistor ($R_L$)
2. A proper load resistor shall be chosen to meet the requirement of maximum ambient light and output saturation voltage:

$$V_{out(max.)} = I_{PH(max.)} \times R_L \leqq V_{out(saturation)} = V_{cc} - 0.4V$$

$$R_L \times C_L \geqq 0.5 \quad \text{(empirical data)}$$


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## Recommended method of storage

1. Do not open moisture proof bag before devices are ready to use.
2. Shelf life in sealed bag from the bag seal date: 18 months at 10°C~30°C and < 90% RH.
3. After opening the package, the devices must be stored at 10°C~30°C and ≤ 60%RH, and used within 168 hours (floor life).
4. If the moisture absorbent material (desiccant material) has faded or unopened bag has exceeded the shelf life or devices (out of bag) have exceeded the floor life, baking treatment is required.
5. If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure or recommend the following conditions:
   192 hours at 40°C +5/–0°C and < 5 % RH (reeled/tubed/loose units) or
   96 hours at 60°C ± 5°C and < 5 % RH (reeled/tubed/loose units) or
   24 hours at 125°C ± 5°C, not suitable for reel or tubes.

## ESD Precaution:

Proper storage and handing procedures should be followed to prevent ESD damage to the devices especially when they are removed from the Anti-static bag. Electro-Static Sensitive Devices warning labels are on the packing.

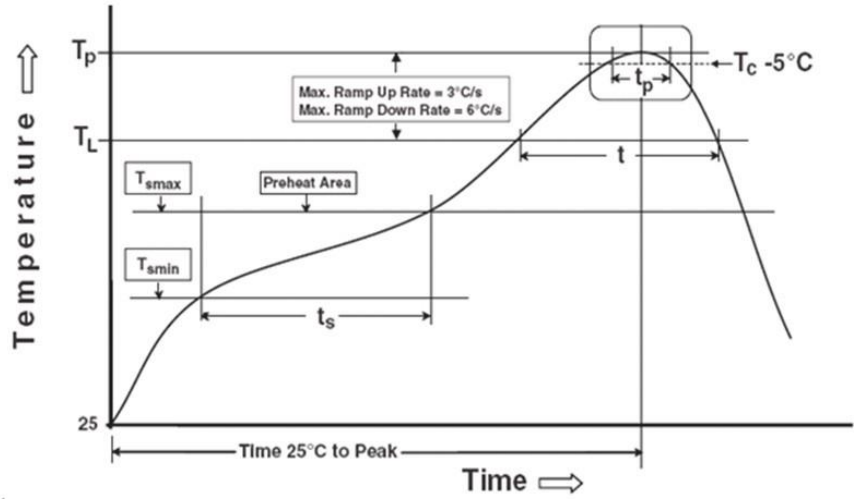
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## Recommended Solder Profile

1. Soldering Condition

1.1 (A) Maximum Body Case Temperature Profile for evaluation of Reflow Profile




**Note:** Reference: IPC/JEDEC J-STD-020D

| | |
|---|---|
| **Preheat** | |
| Temperature min ($T_{smin}$) | 150 °C |
| Temperature max ($T_{smax}$) | 200°C |
| Time ($T_{smin}$ to $T_{smax}$) ($t_s$) | 60-120 seconds |
| Average ramp-up rate ($T_{smax}$ to $T_p$) | 3 °C/second max |
| **Other** | |
| Liquidus Temperature ($T_L$) | 217 °C |
| Time above Liquidus Temperature ($t_L$) | 60-100 sec |
| Peak Temperature ($T_P$) | 260°C |
| Time within 5 °C of Actual Peak Temperature: $T_P$ - 5°C | 30 s |
| Ramp- Down Rate from Peak Temperature | 6°C /second max. |
| Time 25°C to peak temperature | 8 minutes max. |
| Reflow times | 3 times |

**Notice:**

(1) When soldering, do not put stress on the devices during heating.

(2) After soldering, do not warp the circuit board.

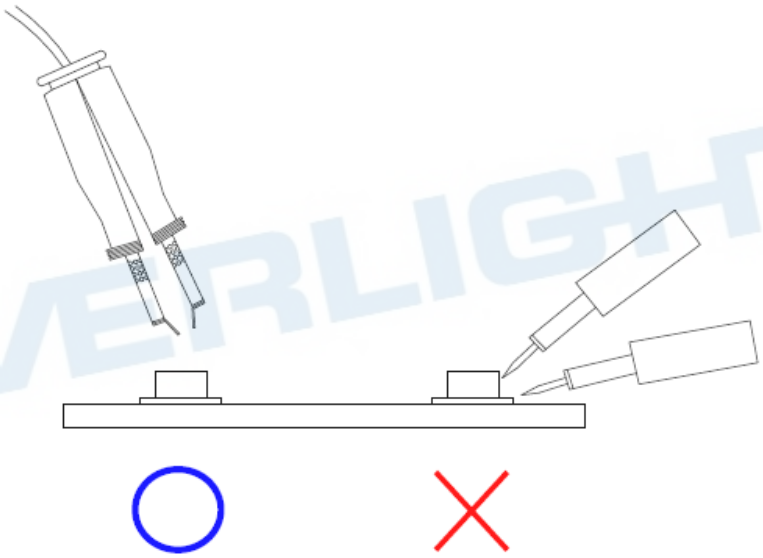

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## Solder Iron

Each terminal is to go to the tip of soldering iron temperature less than 350℃ for 3 seconds within once in less than the soldering iron capacity 25W. Leave two seconds and more intervals, and do soldering of each terminal. Be careful because the damage of the product is often started at the time of the hand solder.

## Repairing

Repair should not be done after the device have been soldered. When repairing is unavoidable, a double-head soldering iron should be used (as below figure). It should be confirmed beforehand whether the characteristics of the device will or will not be damaged by repairing.



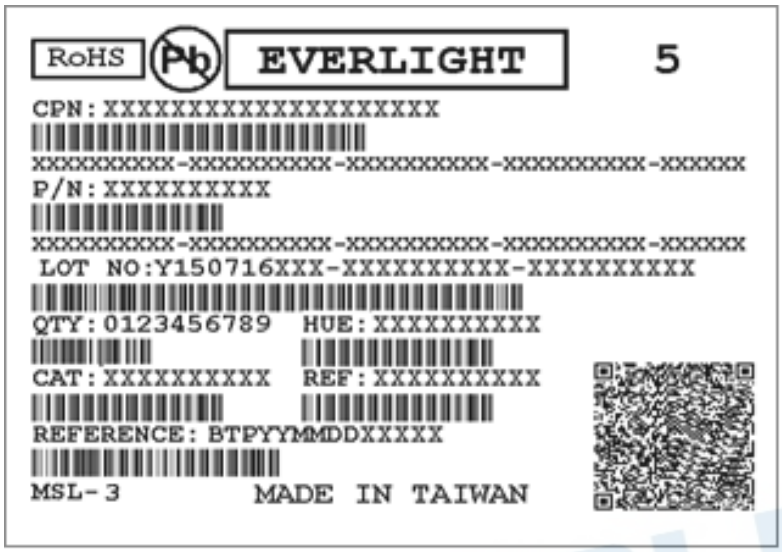

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## Packing Quantity Specification

4000 PCS/ 1 Reel

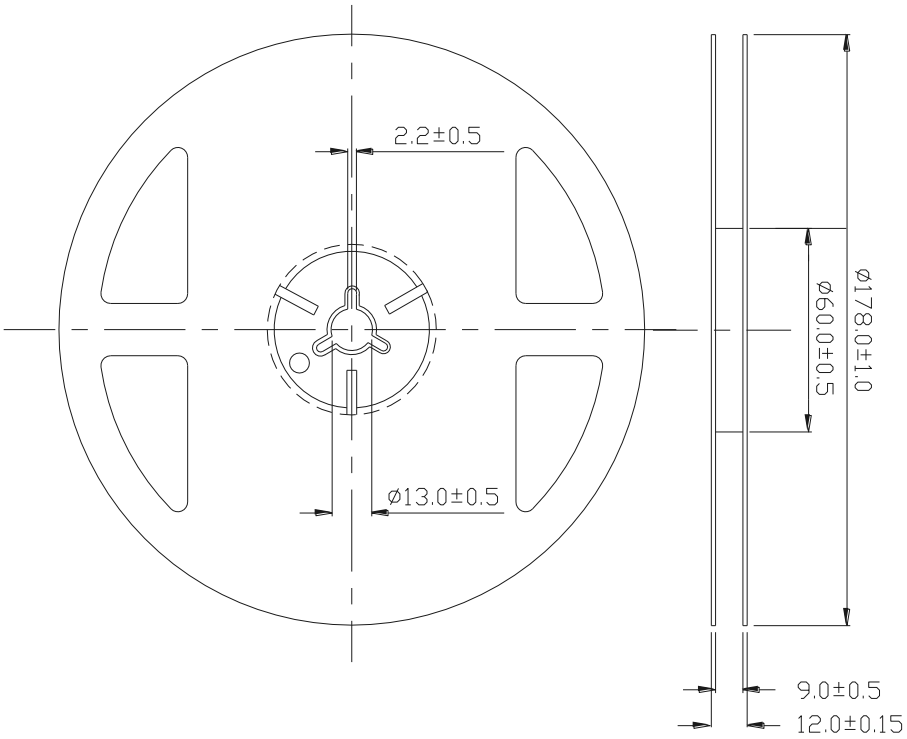
## Label Format




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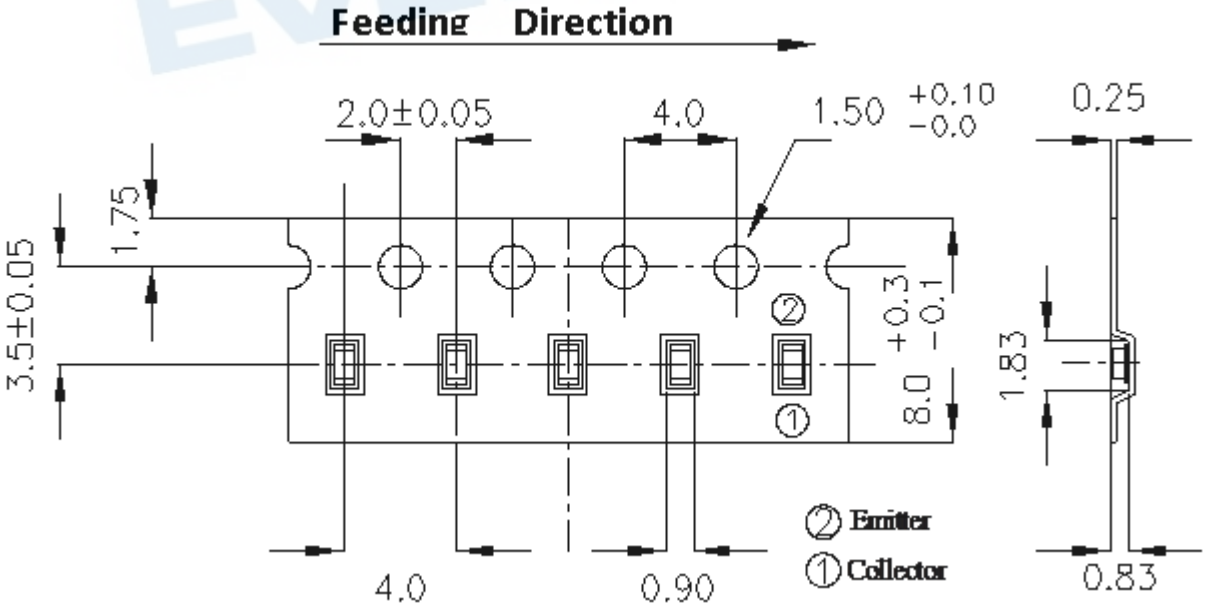

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## Reel Dimensions




Unit: mm
Tolerance: ±0.1mm

## Tape Dimensions




Unit: mm
Tolerance: ±0.1mm


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